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DATAMINING Project
Third Review

Student Academic Performance Prediction & Attendance Impact Analysis

Presented by:

N. Renuka Sri Hasini (24B11A1299)
Anurag Nayak (24B11A1014)

Department of Artificial Intelligence & Machine Learning

Presentation Outline

1. Abstract
2. Introduction
3. Analysis
4. Data Description
5. Preprocessing
6. Algorithms
7. Results
8. Conclusion

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ABSTRACT

- Educational institutions generate large student data
- Attendance and performance are not analyzed together
- Leads to "Attendance-Performance Gap"
- No early prediction system for student failure
- Proposed system uses Data Mining techniques
- Predicts student performance in advance
- Helps in early academic intervention

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INTRODUCTION

- Student data is increasing rapidly
- Includes attendance, marks, assignments
- Traditional systems only store data
- No predictive analysis is done
- Educational Data Mining (EDM) helps extract insights
- Project builds a prediction system for student performance

ANALYSIS

- Reactive approach (after failure)
- Manual data analysis
- No prediction of performance
- Data stored but not utilized
- Decisions based on assumptions

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Limitations of Existing System:

- No early identification of weak students
- Time-consuming manual tracking
- Cannot analyze multiple factors together
- No accurate prediction system
- Delayed academic intervention

Proposed System

- Data-driven prediction system
- Uses student attributes:
  - Attendance
  - Study Hours
  - GPA
  - Assignments
- Applies machine learning algorithms
- Predicts:
  - Fail / Pass / Distinction
  - Provides early warning system

Advantages of Proposed System

- Early identification of at-risk students
- High prediction accuracy
- Reduces manual effort
- Data-driven decision making
- Improves academic performance
- Supports faculty intervention

DATA DESCRIPTION

- Dataset contains student academic data
- Each row = one student
- Attributes include:
  - Attendance Percentage
  - Study Hours
  - GPA
  - Assignments
  - Participation
- Target:
  - Final Grade (Fail / Pass / Distinction)

DATA PREPOCESSING

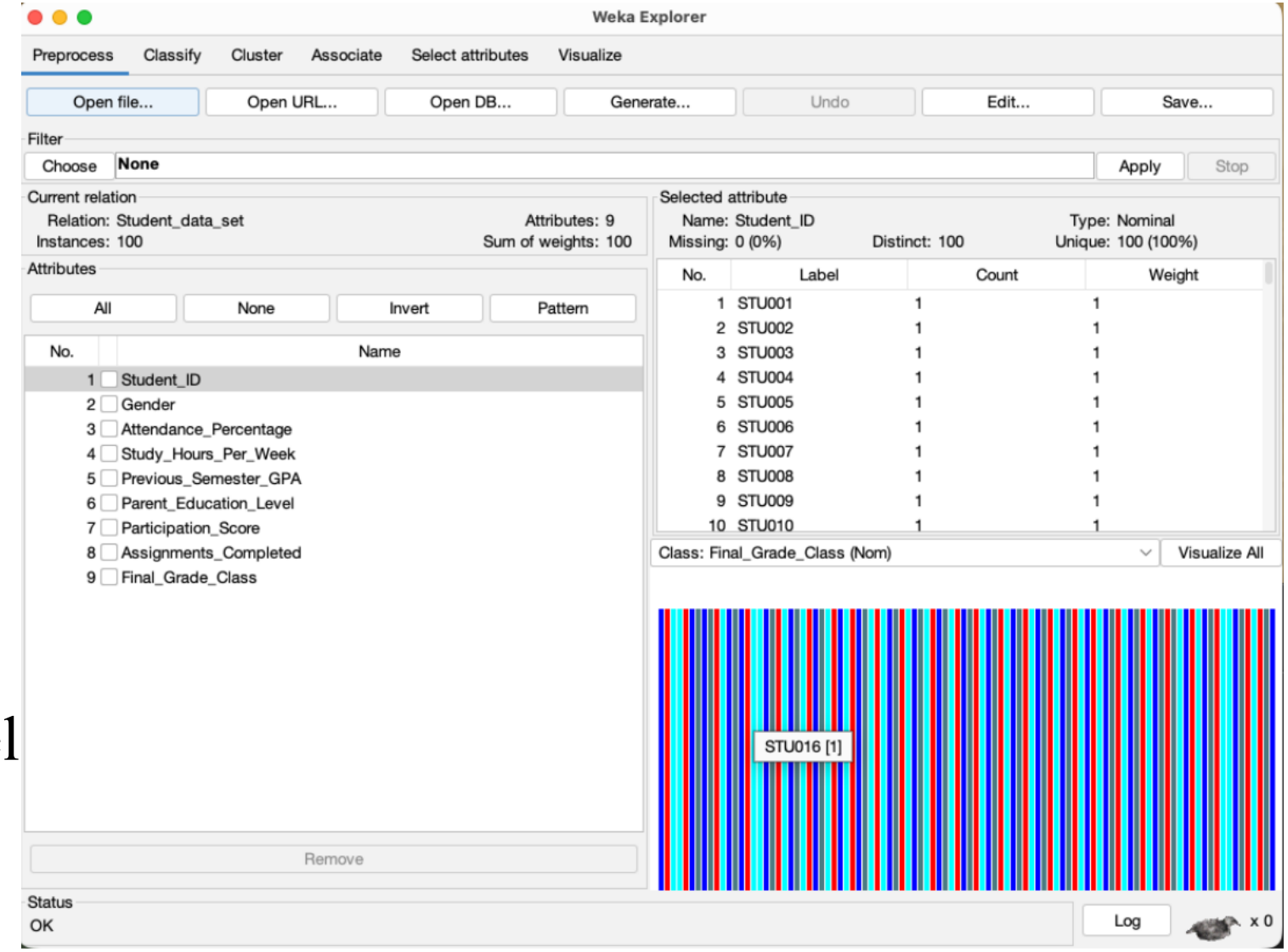
- Raw data is not clean
- Preprocessing improves accuracy

Steps:

1. Removed irrelevant data (Student ID)
2. Handled missing values
3. Converted categorical data
4. Normalized numeric values
5. Removed outliers

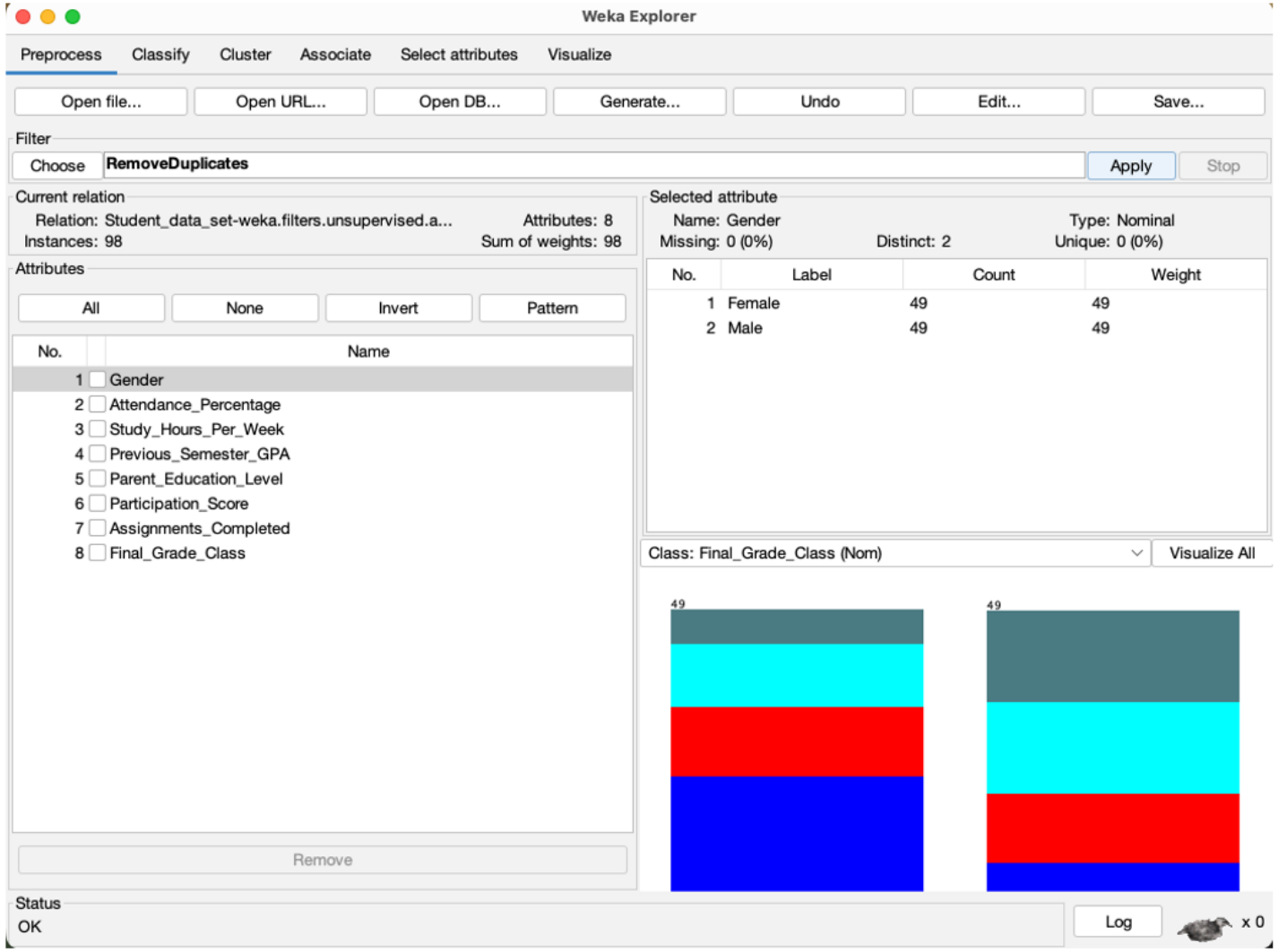
Before Preprocessing

- Missing values present
- Duplicate records exist
- Data not structured
- Not suitable for ML model



After Preprocessing

- Clean dataset
- No missing values
- Structured data
- Normalized values
- Ready for model training

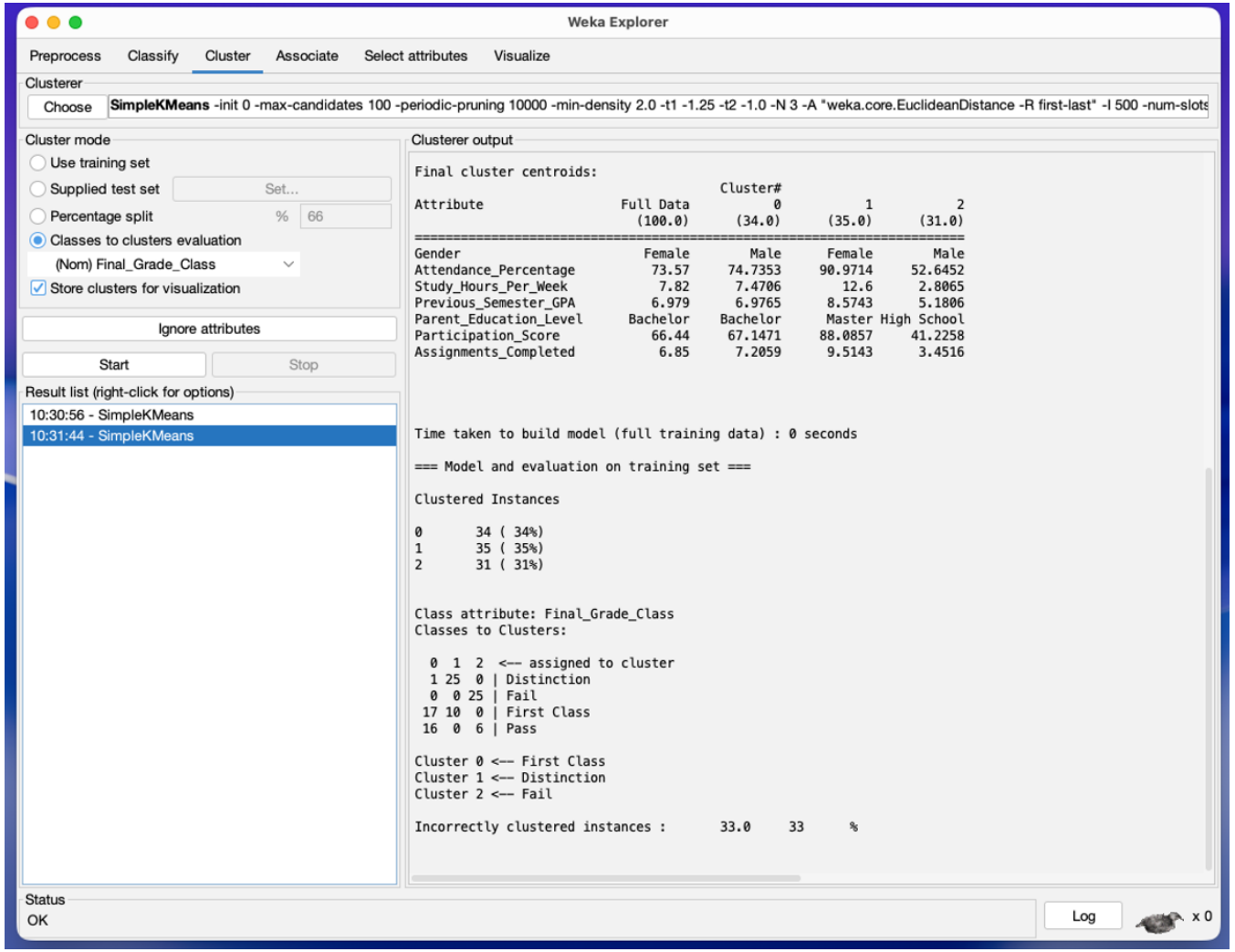


CLUSTERING (K-Means)

- Used for grouping students
- Creates "Learner Personas"

Clusters:

- High Achievers
- Average Students
- At-Risk Students
- Helps identify hidden patterns

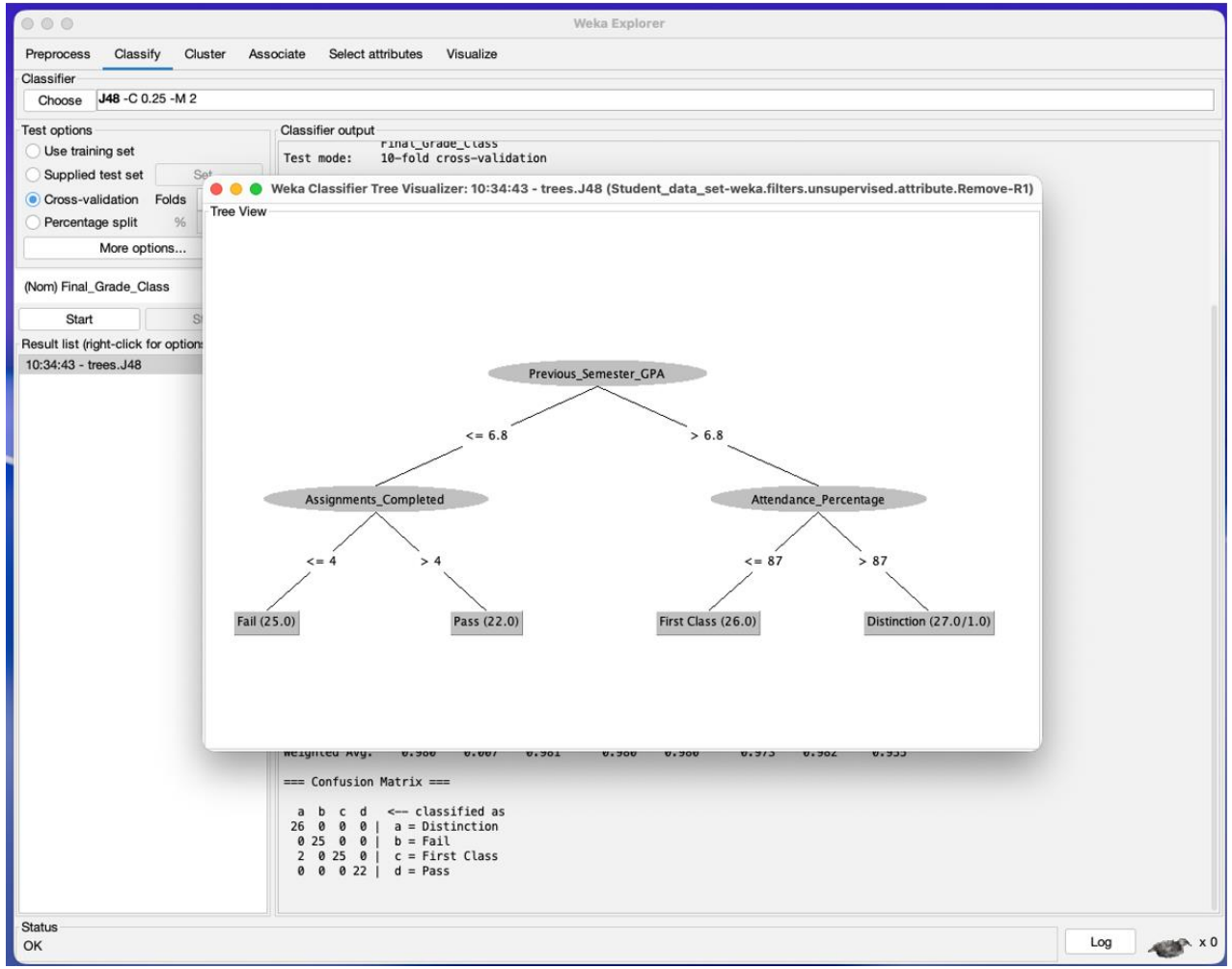


CLASSIFICATION

- Predicts final student performance
- Based on:
  - Attendance
  - GPA
  - Assignments

Output:

- Fail
- Pass
- Distinction
- Easy to understand decision rules

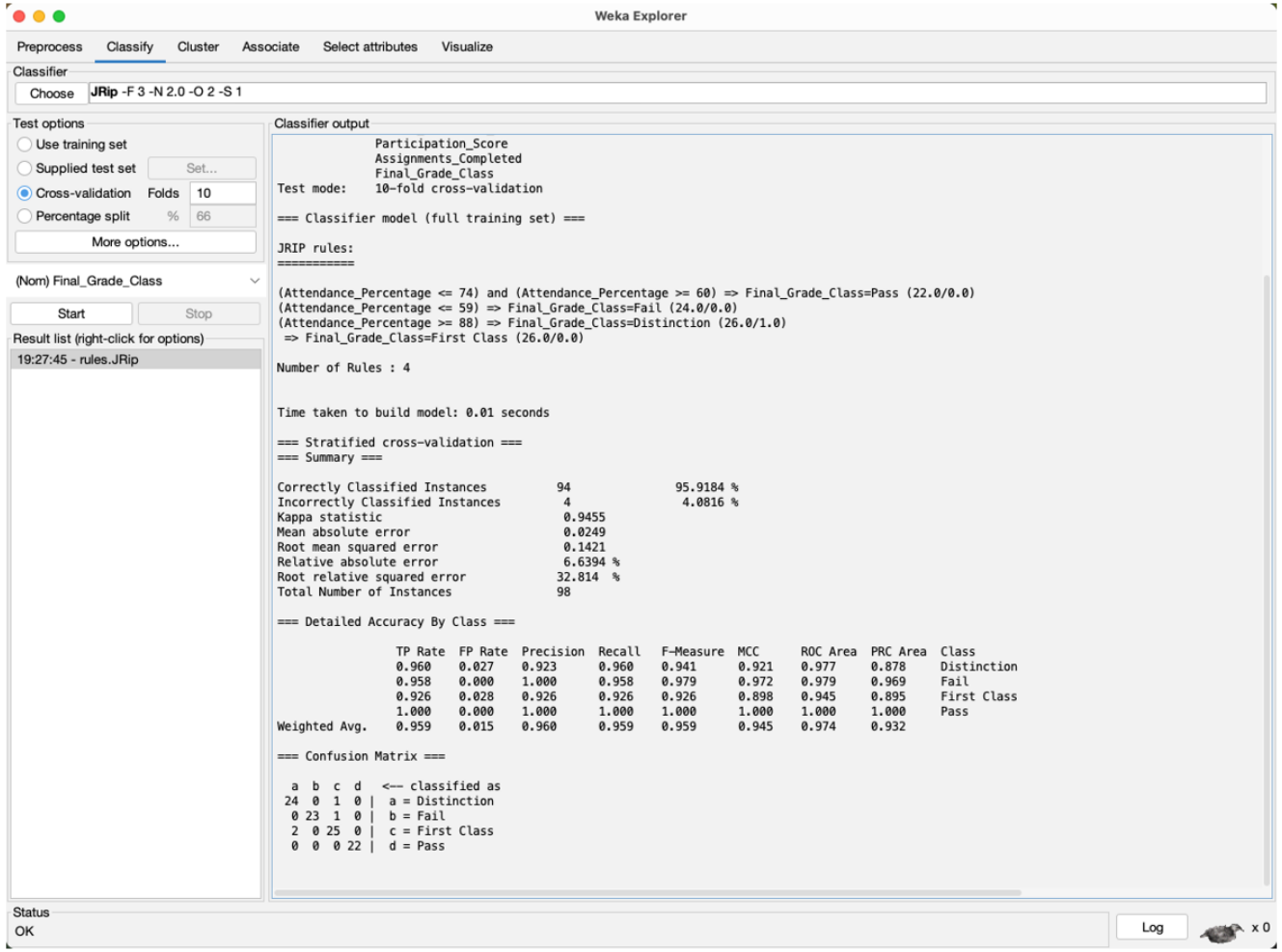


RULE-BASED MODEL (JRip)

- Generates IF-THEN rules
- Easy for faculty to interpret

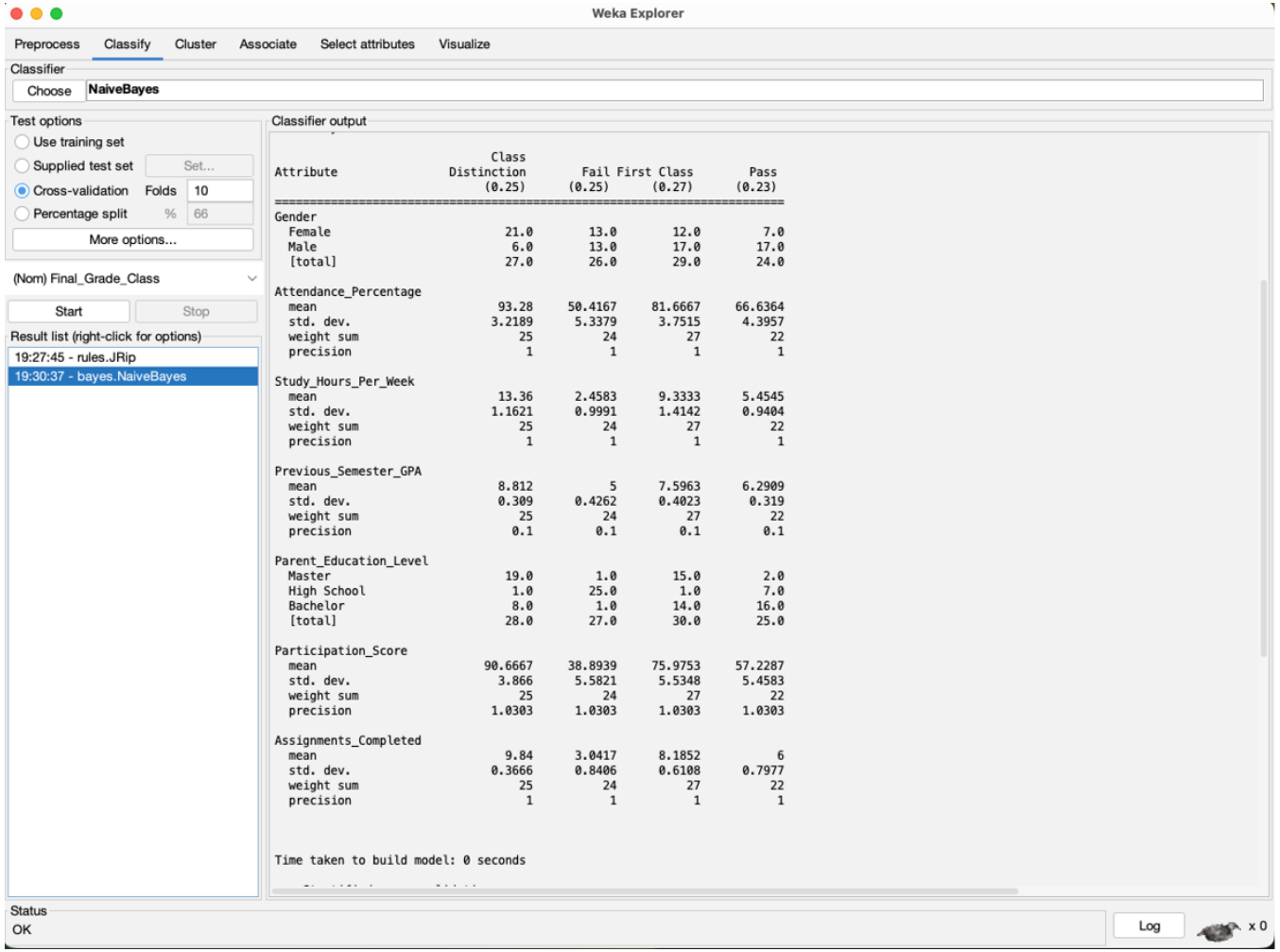
Example:

- IF attendance < 60% → Fail
- IF assignments > 4 → Pass



NAIVE BAYES

- Probabilistic classifier
- Based on Bayes Theorem
- Calculates probability of outcomes
- Fast and simple model



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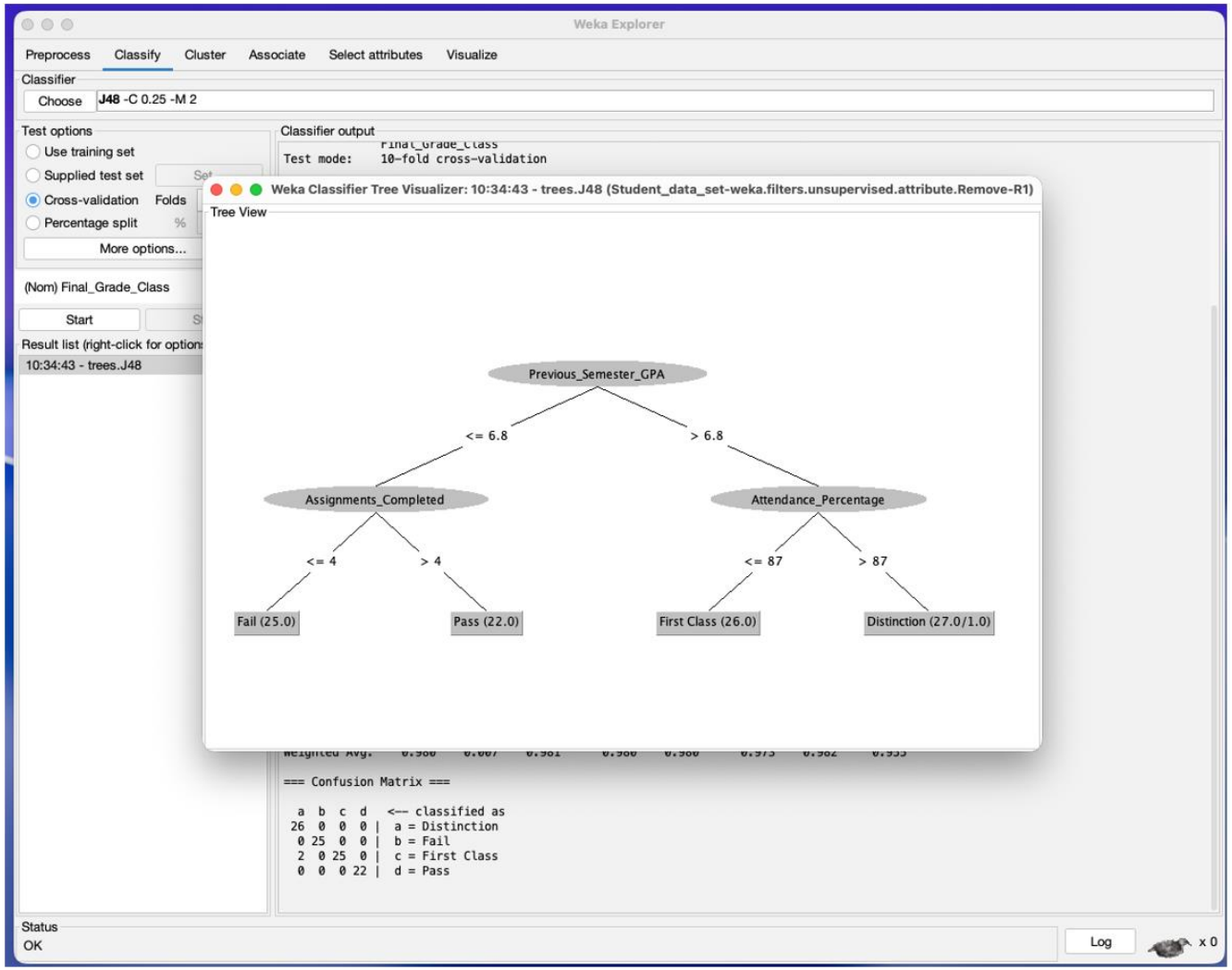
RESULTS (Clustering Output)

- High Achievers:
  - High attendance
  - High study hours
- Average Students:
  - Moderate performance
- At-Risk Students:
  - Low attendance
  - Low study hours

RESULTS (Decision Tree)

- GPA is main factor
- Attendance affects distinction
- Assignments affect pass/fail



MODEL COMPARISON

Model / Algorithm	Correctly Classified Instances (Accuracy)	Incorrectly Classified Instances (Error Rate)	Mean Absolute Error (MAE)	Kappa Statistic	Model Type
J48 Decision Tree	97.96% (96 instances)	2.04% (2 instances)	0.0148	0.9728	Hierarchical Tree
Naive Bayes	95.92% (94 instances)	4.08% (4 instances)	0.0225	0.9455	Probabilistic
JRip (RIPPER)	95.92% (94 instances)	4.08% (4 instances)	0.0249	0.9455	IF-THEN Rule-Based

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CONCLUSION

- Successfully predicts student performance
- Bridges Attendance-Performance Gap
- Helps early identification of weak students
- Improves academic outcomes
- Supports data-driven education system

THANK YOU